

IN VITRO EVALUATION OF THE EFFECT OF POLISHING ON THE COLOUR STABILITY OF IN OFFICE BLEACHING AGENTS FOLLOWING THEIR IMMERSION IN DIFFERENT BEVERAGES

Sae Anawalikar , Harish Selvaraj

Sae Anawalikar

Postgraduate Student

Department of Conservative Dentistry and Endodontics

Saveetha Dental College and Hospital

Chennai

Harish Selvaraj

Senior Lecturer

Department of Conservative Dentistry and Endodontics

Saveetha Dental College and Hospital

Chennai

Corresponding author :

Harish Selvaraj

Senior Lecturer

Department of Conservative Dentistry and Endodontics

Saveetha Dental College and Hospital

Chennai

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ABSTRACT

Objective: To determine the effect of post-bleaching polishing on the color stability of enamel treated with in-office bleaching agents following immersion in different chromogenic beverages.

Materials and Methods: 120 freshly extracted mandibular central incisors with intact, healthy enamel were obtained.

A (n=40) control group.

B. (n = 40) 35 % Hydrogen peroxide (Mani Bleach N Shine Office)

Group B was further subdivided as follows:

B1-(n = 20): 35 % Hydrogen peroxide (Mani Bleach N Shine Office) with polishing using polishing discs (Eve Diacomp polishing kit) and polishing paste (Platina High Gloss)

B2-(n = 20): 35 % Hydrogen peroxide (Mani Bleach N Shine Office) without polishing

C. - (n=40) 40% Hydrogen peroxide (Ultradent Opalescence Boost)

Group C was further subdivided as follows:

C1-(n = 20): 40% Hydrogen peroxide (Ultradent Opalescence Boost) with polishing using polishing discs (Eve Diacomp polishing kit) and polishing paste (Platina High gloss)

C2-(n = 20): 40% Hydrogen peroxide (Ultradent Opalescence Boost) without polishing

10 samples from each group were immersed in coffee solution, and 10 samples from each group were immersed in orange soda solution.

Result:

A statistically significant difference in color change (ΔE) was observed among all groups ($p < 0.05$). The control group exhibited the highest discoloration in both beverages. Specimens bleached with MANI and Ultradent followed by polishing showed significantly lower ΔE values compared to their unpolished counterparts, indicating enhanced color stability. MANI with polishing demonstrated the greatest resistance to staining in coffee, while Ultradent with polishing showed the least discoloration in orange soda. Post-hoc analysis confirmed significant intergroup differences ($p < 0.001$).

Conclusion:

Polishing following the application of in-office bleaching agents significantly enhances color stability when the treated enamel is subsequently exposed to chromogenic beverages.

Keywords: In Office Bleaching, colour stability, polishing agents, healthy lifestyle, medicine

INTRODUCTION

The pursuit of aesthetically pleasing, whiter smiles has significantly contributed to the growing popularity of in-office bleaching as a central modality in contemporary cosmetic dentistry. This technique is favored for its ability to deliver rapid and visibly effective results, thereby providing immediate patient satisfaction^[1]. The primary active agents employed in in-office bleaching are hydrogen peroxide and carbamide peroxide. These low-molecular-weight compounds readily diffuse through dental tissues, initiating oxidative reactions that degrade chromogenic macromolecules into smaller, colorless byproducts, ultimately leading to tooth whitening^[2].

Despite the proven efficacy of these bleaching agents, concerns persist regarding their potential effects on dental hard tissues^[3]. Studies have reported several structural and chemical alterations in enamel following bleaching, including increased surface roughness, decreased microhardness, modifications in chemical composition, and compromised bond strength between bleached enamel and resin-based materials^[4]. Additionally, peroxide-induced changes in the organic matrix and protein layer of the enamel may enhance its permeability, rendering the surface more susceptible to extrinsic staining from dietary chromogens^[5]. Consequently, the microstructural alterations induced by bleaching are considered critical factors that may undermine the longevity and aesthetic stability of the treatment^[6].

Post-bleaching, the increased porosity of enamel facilitates the penetration of chromogenic substances from commonly consumed beverages and foods. Several investigations suggest that immediate consumption of pigmented dietary items following bleaching can exacerbate staining and reverse the whitening effect^[7]. Therefore, strategies to mitigate post-bleaching discoloration are essential to enhance treatment outcomes.

Polishing the enamel surface after bleaching has been proposed as a potential intervention to counteract the deleterious effects of peroxide exposure. By smoothing the surface and reducing porosity, polishing may limit pigment retention and enhance color stability^[8]. However, limited literature exists on the role of polishing as a preventive measure against post-bleaching discoloration. Hence, the present in vitro study aimed to evaluate the effect of post-bleaching polishing on the color stability of enamel surfaces exposed to common staining beverages.

Null Hypothesis

1. Post-bleaching tooth polishing has no significant effect on the color stability of enamel.
2. There is no significant difference in color stability between teeth immersed in different chromogenic beverages following in-office bleaching.

Aim of the Study

To evaluate the influence of post-bleaching polishing on the color stability of in-office bleaching agents.

Objectives of the Study

Primary Objective:

To determine the effect of polishing on the color stability of enamel following in-office bleaching procedures.

Secondary Objective:

To compare the impact of different chromogenic beverages on the color stability of enamel treated with in-office bleaching agents.

MATERIALS AND METHODOLOGY

SAMPLE PREPARATION

Study Design :

The study was conducted in the Department of Conservative Dentistry and Endodontics at a university-affiliated dental research laboratory. Ethical clearance was obtained from the Institutional Human Ethics Committee (Approval No. SRB/SDC/ENDO-2307/24/071), under the ethical standards of the Helsinki Declaration and its subsequent amendments. The methodology of the study was conducted as follows

G*POWER Version 3.1.9.7 was used to calculate the sample size. 120 freshly extracted mandibular central incisors with intact, healthy enamel were obtained. The teeth were evaluated under a stereoscopic magnifying glass (40X) to check the surface integrity for absence of enamel cracks, fractures, and defects. The teeth showing any of the defects were excluded from the study.

SAMPLE ALLOCATION

The teeth were categorized into distinct groups through a process of simple randomization. All procedures were conducted by a single operator, while the evaluation of the samples was performed by a separate blinded investigator. The groups were classified as follows

A. (n=40) control group.

B-(n= 40) 35 % Hydrogen peroxide (Mani Bleach N Shine Office)

Group B was further subdivided as follows:

B1- (n=20): 35 % Hydrogen peroxide (Mani Bleach N Shine Office) with polishing using Diacomp polishing discs and Platina High gloss Polishing paste

B2- (n=20): 35 % Hydrogen peroxide (Mani Bleach N Shine Office) without polishing

C- (n=40) 40% Hydrogen peroxide (Ultradent Opalescence Boost)

Group C was further subdivided as follows:

C1- (n =20): 40% Hydrogen peroxide (Ultradent Opalescence Boost) with polishing using Diacomp polishing discs and Platina High gloss Polishing paste

C2- (n = 20): 40% Hydrogen peroxide (Ultradent Opalescence Boost) without polishing

10 samples from each group were immersed in coffee solution, and 10 samples from each group were immersed in orange soda solution.

Table 1: List of materials used for bleaching, polishing and staining

Product	Commercial Name
35% Hydrogen Peroxide	Bleach'n Shine Bleach Office, Mani, Inc.
Polishing discs	Diacomp
Polishing Paste	Platina High Gloss
Coffee Solution	Bru Instant Aromatic Coffee
Orange Soda Solution	Mirinda

Before bleaching, biofilm removal was done for all the groups using ultrasonic scaler and Ultrasonic Scaler Perio Scaling Tip P3 (Woodpecker)

BLEACHING PROCEDURE:

Bleaching procedures were conducted for Group B utilizing 35% hydrogen peroxide (Mani Bleach N Shine Office) and for Group C employing 40% hydrogen peroxide (Ultradent Opalescence Boost). The gel was applied in accordance with the manufacturer's instructions, allowing for a 20-minute application per session over the course of three sessions, followed by rinsing with distilled water. The bleaching sessions were scheduled at intervals of seven days, during which the samples were maintained in artificial saliva. Following each bleaching session, polishing was performed using Diacomp polishing discs and Platina High Gloss polishing paste at a speed of 15,000 rpm for a duration of 5 to 10 seconds.

IMMERSION IN BEVERAGES

Following the three bleaching treatments, 10 samples from each group were immersed in the coffee solution and 10 samples in the orange soda solution. To accurately imitate oral circumstances, the following approach was used: 1) soak for 5 minutes in 3 mL of the coloring solution (coffee, orange soft drink); 2) rinse with 3 mL of distilled water; and 3) store in artificial saliva for 24 hours.

This cycle was performed four times a day for twenty-one days.^[7]

COLOUR MEASUREMENT:

The color measurement for all the specimens was done immediately after the last session of bleaching before the staining procedure, using a digital spectrophotometer.

The color measurement was done after the last immersion cycle of the specimens.

The difference in color was calculated.

RESULTS :

Color Stability Following Immersion in Beverages

A statistically significant difference in color change (ΔE values) was observed among the groups immersed in coffee and orange soda ($p < 0.05$). The control group (unbleached specimens) exhibited the highest discoloration, with a mean ΔE of 8.688 ± 0.092 in coffee and 2.408 ± 0.135 in orange soda.

Among the treated groups:

- MANI bleaching agent followed by polishing demonstrated the lowest ΔE values, indicating superior color stability ($\Delta E = 2.570 \pm 0.089$ in coffee and 1.872 ± 0.048 in orange soda).
- Ultradent bleaching agent followed by polishing also exhibited improved stability ($\Delta E = 5.468 \pm 0.101$ in coffee and 1.260 ± 0.038 in orange soda), with lower discoloration compared to its corresponding unpolished group.
- Specimens treated with MANI or Ultradent without polishing showed higher ΔE values than their polished counterparts, indicating that polishing post-bleaching reduced susceptibility to staining.

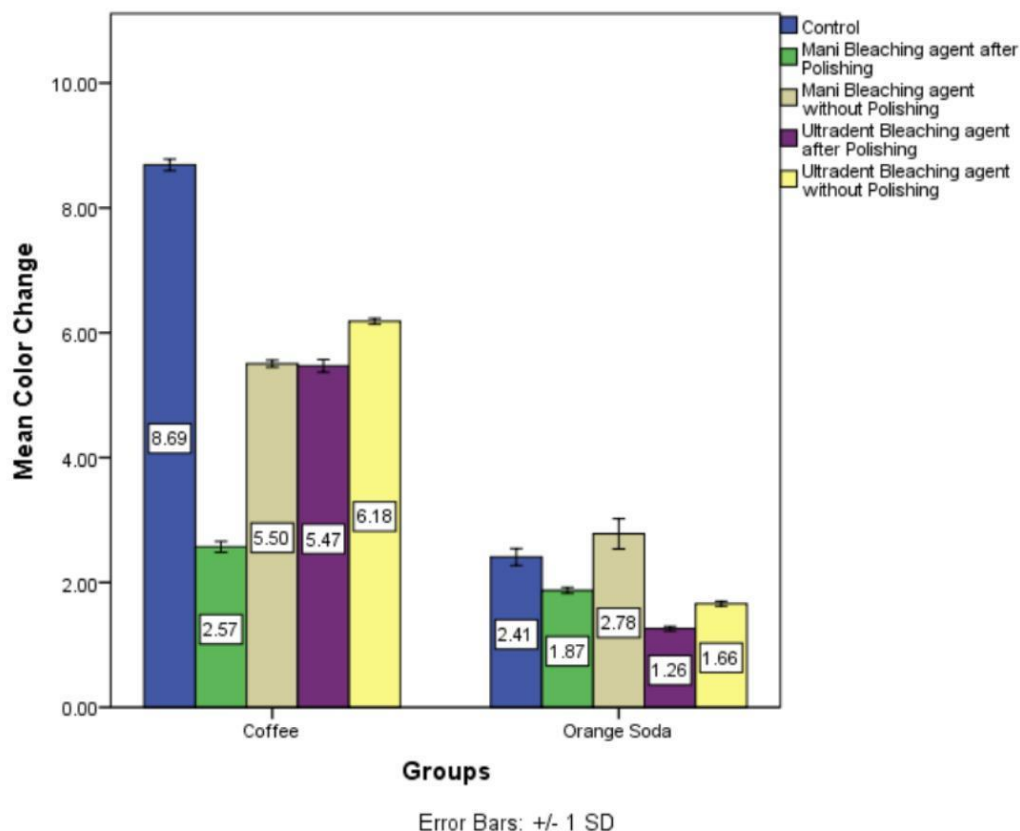


Figure 1: Bar graph depicting the mean color change (ΔE) of dental specimens subjected to staining with coffee and orange soda following in-office bleaching using two agents—Mani and Ultradent—with and without polishing

Independent Samples t-test

Statistical comparison with the control revealed that all bleached groups showed significantly lower ΔE values ($p < 0.001$). The greatest mean difference was recorded between the control and the MANI post-polishing group (mean difference = 6.28, 95% CI: 6.17–6.39). Ultradent post-polishing also showed a significant difference compared to the control (mean difference = 4.208, $p < 0.001$).

Post-hoc Analysis (Tukey HSD)

In the coffee immersion subgroup, the MANI post-polishing group had significantly lower discoloration than all other groups ($p < 0.001$). Both MANI and Ultradent groups without polishing differed significantly from the control as well as from each other ($p < 0.001$).

In the orange soda subgroup, Ultradent post-polishing showed the least color change ($\Delta E = 1.260$), which was significantly lower than all other groups ($p < 0.001$). The MANI post-polishing group also exhibited significantly reduced ΔE values compared to unpolished groups and the control ($p < 0.001$).

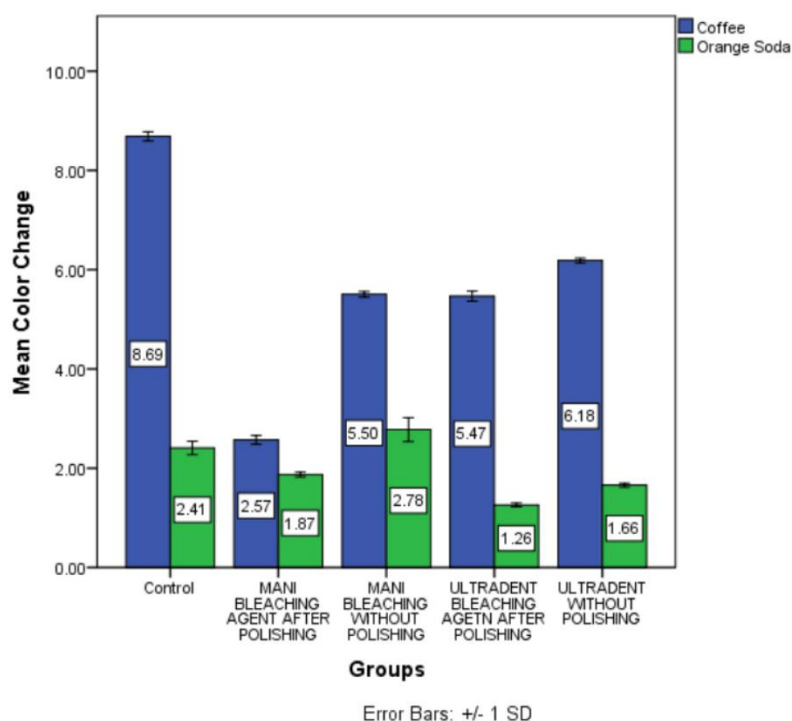


Figure 2: Bar graph representing the mean color change (ΔE) of specimens subjected to staining with coffee and orange soda after in-office bleaching using two agents (Mani and Ultradent), with and without polishing.

DISCUSSION

The present in vitro study aimed to evaluate the effect of polishing on the color stability of commercially available in-office bleaching agents, with the objectives to evaluate the effect of polishing on the color stability of in-office bleaching agents and the difference in the effect of different pigmented substances on the color stability. 60 mandibular central incisors with intact enamel were obtained and divided into groups, including a control group, a group with 35% hydrogen peroxide (Mani Bleach N Shine Office), a group with polishing using Diacomp discs and Platina High Gloss Polishing Paste, and a group with 40% hydrogen peroxide (Ultradent Opalescence Boost).

According to Nogueira et al., consuming colored drinks while receiving bleaching therapy can slow down the process. This surface is more easily stained. Côrtes et al. observed discoloration of teeth during and after whitening with coffee and red wine. Artificial saliva effectively reduced enamel discoloration by enabling enamel remineralization during therapy. Nonetheless, after the bleaching, both the coffee and red wine stain treatments caused changes in the color of the enamel. Additionally, these beverages have the potential to produce discoloration in composite resins.^[9]

The results indicated that polishing has an important role in the color stability of the bleached teeth, even after immersion in the staining agents. In-office bleaching procedures, though effective in beautifying teeth, discolor a few minutes after treatment when teeth come into contact with chromogenic beverages.^{[10],[11]} Coffee and orange juice have been shown in previous research as

the largest contributors of extrinsic stains due to their chromogenic nature and low pH, which could impact the enamel surface and cause color changes. ^[12]

This protective role that polishing plays in this study can be attributed to smoothening the enamel surface, thereby reducing surface roughness, which would normally help pigment adhesion from beverages. ^[13] Possibly, polishing minimizes the micro-porosities that trap stains and, with findings from other studies, smoother enamel surfaces are less prone to discoloration post-bleaching. ^{[14],[15]}

Although polishing generally showed a beneficial effect on color stability, the protective effect was very different for beverages. The potential staining of coffee, a rich source of tannins and chromogens, was more significant compared with orange juice; this is in partial agreement with previous studies. ^{[16],[17]} Nevertheless, the acidity of orange juice could take a toll on the enamel and so may increase the likelihood of staining over time because of erosion if not balanced with protective methods like polishing. ^{[18],[19]} In our study, we found that 40% Hydrogen Peroxide with polishing was highly effective in reducing the Change in Color Stability. Our study shows that adding polishing to the post-bleaching regime significantly extends the period of the aesthetic effects. ^{[20],[21]} Further studies are needed to investigate the lasting effects of the various techniques concerning their resistance to different dietary challenges. ^{[22],[23]}

CONCLUSION

In conclusion, the study found that the color stability of samples varied depending on the treatment and exposure to coffee or orange juice immersion. The group treated with a 40% concentration of hydrogen peroxide (H₂O₂) and a polishing agent in orange juice had the least color change. ^[24]

The control group that had orange juice had the highest color change. The statistical analysis

showed significant differences in color stability among all the different groups. Treatments with polishing generally showed better color stability than treatments without polishing. Immersion in coffee generally caused more significant color change than immersion in orange juice.^{[25],[26]}

AUTHOR CONTRIBUTION

CONFLICT OF INTEREST

The authors declare no conflict of interest

AUTHORS CONTRIBUTION

SA,PS conceptualized the study and supervised the project. SA performed the experimental work and collected data. PS,HS conducted statistical analysis and interpreted the results. SA,SS,PS prepared the manuscript draft. All authors reviewed and approved the final manuscript.

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